

used. To know how much current a given wire can carry, SMPS designers use a number called current density [J] for this calculation. Empirically, a good starting point is  $J = 5 \text{ amps/mm}^2$ . The primary carries 0.3 amps, so it can be wound using SWG 38. The secondary carries 5 amps, so it can be wound with SWG 26. A good practice would be to wind the secondary using two parallel strands of SWG 28. This reduces the skin effect. The key problem in flyback transformers is leakage inductance, which is caused by poor coupling between the primary and secondary windings. So wind them tight, with full overlap.

**The air gap.** Flybacks made from power ferrites must have an air gap. The energy stored in a flyback primary is  $E = \frac{1}{2} L I^2$ . Peak primary current depends inversely on primary inductance. An air gap increases the energy storage capacity of a flyback transformer. It is calculated as follows:

$$\text{Air gap} = (\mu_0 \times N^2 \times A_c) / L_{\text{primary}}$$

where:

$$\mu_0 = \text{permeability of free space, } 4\pi \times 10^{-7}$$

$$A_c = \text{Core area (m}^2\text{)}$$

$$N = \text{Primary turns}$$

$$L_{\text{primary}} = (V_{\text{primary.pk}} \times T_{\text{on.primary}}) / I_{\text{primary.pk}}$$

The primary inductance computes to ~3mH, so the air gap is approximately 0.2mm. However, this value is not critical, as was experienced by the authors during their experiments. Any air gap in the neighbourhood of the calculated value works well. A thin sheet of plastic or paper works fine.

An actual-size, single-side PCB for the simple 12V, 1A SMPS is shown in Fig. 3 and its component layout is shown in Fig. 4. Assemble the circuit on the recommended PCB to minimise assembly errors. Use IC base for IC1.

To test if the circuit is functioning properly, first check the regulated output at TP1 with respect to TP0. The

## PARTS LIST

## Semiconductors:

|     |                             |
|-----|-----------------------------|
| IC1 | - TNY266 tiny switch-II     |
| IC2 | - EL817 opto-coupler        |
| D1  | - 1N4007 diode              |
| D2  | - P6KE200A TVS diode        |
| D3  | - FR106 fast recovery diode |
| D4  | - SB160 Schottky diode      |
| ZD1 | - 1W, 1W zener diode        |

Resistors (all  $\frac{1}{4}$ -watt,  $\pm 5\%$  carbon, unless stated otherwise):

|    |                |
|----|----------------|
| R1 | - 680-kilo-ohm |
| R2 | - 20-ohm, 0.5W |
| R3 | - 1-kilo-ohm   |

## Capacitors:

|    |                                 |
|----|---------------------------------|
| C1 | - 10 $\mu$ F, 400V electrolytic |
| C2 | - 470 $\mu$ F, 25V electrolytic |
| C3 | - 0.1 $\mu$ F, 400V polyester   |

## Miscellaneous:

|      |                                 |
|------|---------------------------------|
| CON1 | - 5mm terminal connector        |
| CON2 | - 2-pin connector               |
| NL1  | - Neon lamp                     |
| X1   | - EE20 core Flyback transformer |

voltage should be stable with or without a load. ●

Ashvini Vishvakarma is associated with IBM as a telecom systems architect and a radio amateur. Atanu Dasgupta was a powergrid executive in telecom division and is also a radio amateur



www.pcbpower.com®

IDEALAB

## Online PCB Specialist

- PCBs Upto 24 Layers
- No tooling charges
- No minimum order value
- Flexible Deliveries from 24 hours onwards
- Online Ordering for new and repeat orders
- Fast and Reliable logistics
- Online Price Calculator
- Minimum order quantity 1



## Prototype to Production

- Power Express
- Power Verified
- Power Demand

## Industries

- ✓ Automotive
- ✓ Industrial
- ✓ R & D
- ✓ Railway
- ✓ Telecom
- ✓ Space

Call us @ 7600012414 & 7600012415  
E-mail - sales@pcbpower.com

Circuit Systems India Ltd.  
B-24, GIDC, Electronics Estate, Sector-25,  
Gandhinagar-382044, Gujarat, India